

Lab: Case Study - Review a Product Concept

Product-Manager Deliverable for WonderMiner 3000 | Written by Brian McCarthy

Scope Area	Definition
Product concept	WonderMiner 3000 - autonomous mining robot for safer, more productive, and more sustainable mining operations.
Primary users	Mining operators, site managers, remote-control center staff, maintenance teams, safety teams, and mining-company executives.
Business goal	Improve productivity, reduce worker exposure to hazardous conditions, lower transportation and processing waste, and differentiate through AI-enabled automation and real-time operational insight.
Project outcome	A validated product plan, MVP feature set, team structure, RACI chart, and risk/challenge list that guide development and launch readiness.

Step 1 - Skills Required for the Product Manager

Skill	Why it matters for WonderMiner 3000
Market discovery and customer research	Ability to interview mining operators, safety leaders, engineers, executives, and field technicians to validate the highest-value problems.
Technical product judgment	Working knowledge of robotics, AI, sensors, autonomous navigation, mining workflows, data platforms, hardware constraints, and industrial safety requirements.
Product strategy and prioritization	Ability to define the product vision, MVP, roadmap, success metrics, release sequencing, and trade-offs across safety, cost, feasibility, and customer value.
Stakeholder management	Ability to align engineering, operations, safety, legal, marketing, sales, finance, and executive sponsors around a clear product direction.
Risk and compliance management	Understanding of operational risk, environmental impact, worker safety, regulatory expectations, cybersecurity, and remote-control reliability.
Data-driven decision-making	Ability to use telemetry, market research, pilot results, adoption data, safety metrics, and operating-cost data to guide product decisions.
Go-to-market thinking	Ability to support positioning, pricing, pilot-customer selection, training materials, buyer messaging, and launch-readiness planning.

Step 2 - Key Tasks / Activities

#	Activity	Description
1	Clarify product vision and success metrics	Define the vision, target customers, measurable outcomes, launch criteria, and business objectives for the WonderMiner 3000.
2	Conduct market, user, and field research	Interview mining operators, safety managers, buyers, and technicians; study competitor offerings; observe site workflows and constraints.
3	Validate feasibility with engineering	Confirm robotics, AI, sensor, onboard processing, battery/power, ruggedization, and remote-control feasibility.
4	Define MVP and prioritize features	Prioritize autonomous navigation, hazard detection, remote monitoring, telemetry, extraction modules, and onboard processing by value and risk.
5	Create requirements and acceptance criteria	Translate customer needs into product requirements, user stories, safety requirements, and measurable acceptance criteria.
6	Plan pilot program and field testing	Select pilot sites, define test scenarios, collect telemetry, assess safety performance, and validate operational productivity.
7	Coordinate compliance, safety, and environmental review	Review safety protocols, emergency shutdown, hazardous gas detection, terrain risks, operator training, and environmental impact.
8	Develop launch and go-to-market plan	Coordinate pricing input, sales enablement, training, implementation support, marketing positioning, and launch communications.
9	Measure post-launch outcomes	Track adoption, uptime, productivity, safety incidents, extraction quality, customer satisfaction, and roadmap feedback.

Step 3 - Product Management Team Roles

Role	Contribution
Product Manager	Owns product vision, customer value, prioritization, roadmap, requirements, launch readiness, and product success metrics.
Engineering Lead	Owns technical solution design, architecture, robotics/AI feasibility, engineering estimates, and development execution.
UX / Industrial Designer	Designs operator experience, control-center workflows, alerts, dashboards, and safe human-machine interaction.
Safety & Compliance Lead	Guides safety protocols, regulatory review, hazard detection requirements, environmental controls, and field operating procedures.
Data / AI Lead	Owns machine-learning models, sensor data analysis, telemetry, optimization algorithms, and model performance monitoring.
Operations / Field SME	Provides mining-site expertise, pilot-site feedback, maintenance knowledge, and real-world operating constraints.
Marketing / Sales Lead	Defines positioning, buyer messaging, sales enablement, pilot customer pipeline, pricing input, and market launch support.
Executive Sponsor	Approves strategic direction, funding, major scope trade-offs, and cross-functional escalation decisions.

Step 4 - RACI Matrix

R = Responsible | A = Accountable | C = Consulted | I = Informed. Each activity has one accountable owner.

Activity	Product Manager	Engineering Lead	UX / Industrial Designer	Safety & Compliance	Data / AI Lead	Operations / Field SME	Marketing / Sales	Executive Sponsor
Clarify product vision and success metrics	A/R	C	C	C	C	C	C	I
Conduct market, user, and field research	A/R	C	C	C	I	R/C	C	I
Validate technical and operational feasibility	A	R	C	C	R/C	C	I	I
Define MVP and prioritize roadmap	A/R	C	C	C	C	C	C	I
Write requirements and acceptance criteria	A/R	C	C	C	C	C	I	I
Design operator workflows and dashboard experience	A	C	R	C	C	C	I	I
Plan safety, compliance, and environmental controls	A	C	I	R	C	C	I	I
Plan pilot program and field tests	A/R	C	I	C	C	R	I	I
Prepare launch and go-to-market plan	A	I	I	C	I	C	R	C
Measure post-launch outcomes and roadmap feedback	A/R	C	I	C	R/C	C	C	I

Potential Challenges the Product Manager May Face

Challenge	Product-management impact
Technical uncertainty	Autonomous navigation, AI decision-making, sensor reliability, terrain adaptation, and onboard processing may be difficult to deliver safely in unpredictable mining environments.
Safety and liability risk	Mining conditions can include gas leaks, unstable terrain, equipment failures, and hazardous materials, so product decisions must prioritize operator and environmental safety.
Regulatory and environmental complexity	Mining regulations, environmental rules, data/security requirements, and site-specific operating constraints may vary by region and customer.
Cross-functional alignment	Engineering, safety, operations, marketing, sales, and executives may prioritize different outcomes, requiring clear trade-off decisions and strong governance.
Customer adoption and trust	Mining customers may be cautious about replacing or supplementing human-operated equipment with autonomous robotics, especially before field results are proven.
Cost and ROI pressure	The product may require expensive hardware, sensors, field support, maintenance, and training, so buyers will expect a clear productivity and safety return.
Pilot execution risk	Field pilots may expose reliability, maintenance, connectivity, training, or workflow issues that require roadmap changes before scale-up.

Recommended Product-Management Practices

Recommendation	Action
Use discovery before delivery	Validate mining-site pain points, buyer priorities, and safety needs before locking the MVP.
Run staged pilots	Use controlled pilots to test autonomy, safety, extraction performance, remote monitoring, and maintenance before broader launch.
Maintain a prioritized roadmap	Keep roadmap decisions tied to customer value, risk reduction, technical feasibility, and measurable business outcomes.
Define clear launch metrics	Track uptime, safety incidents, extraction throughput, material quality, processing efficiency, operator adoption, and customer satisfaction.
Create a governance cadence	Use recurring product reviews with engineering, safety, field SMEs, marketing, and executives to resolve trade-offs quickly.